# PG No. 12 $D_4$ 422 [ tetragonal ]

* generator : $2_{001},\ 4^+_{001},\ 2_{010}$
* conjugacy class

$[1]$ : 1
$[2_{001}]$ : $2_{001}$
$[2_{100}]$ : $2_{100},\ 2_{010}$
$[2_{110}]$ : $2_{110},\ 2_{1-10}$
$[4^+_{001}]$ : $4^+_{001},\ 4^-_{001}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | 1 | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 2 | $2_{001}$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 3 | $4^+_{001}$ | $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 4 | $4^-_{001}$ | $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 5 | $2_{010}$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | 1 |
| 6 | $2_{100}$ | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | 1 |
| 7 | $2_{110}$ | $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | 1 |
| 8 | $2_{1-10}$ | $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ | 1 |